Predictive Processing & Active Inference

A Comprehensive Mathematical & Conceptual Guide

The Free Energy Principle, the Bayesian Brain,
and Consciousness as Controlled Hallucination

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Free Energy • Generative Models • Markov Blankets • Precision • Selfhood

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Chapter 1

Introduction to Predictive Processing

1.1 What is Predictive Processing?

Predictive Processing (PP) is a theoretical framework proposing that the brain is fundamentally a **prediction machine**. Rather than passively receiving and processing sensory input bottom-up, the brain actively generates **top-down predictions** about what it expects to sense, and then compares these predictions with actual sensory input. The difference—the **prediction error**—drives learning, perception, and action.

Core Claim of Predictive Processing

The brain maintains a **generative model** of the world—a hierarchical, probabilistic model that generates predictions about sensory input at every level of the cortical hierarchy. Perception is not a passive reading of sensory data but a process of **Bayesian inference**: the brain infers the most probable causes of its sensory signals by iteratively updating its generative model to minimize **prediction error**. Consciousness arises from the brain's "best guess" about the hidden causes of its sensations.

The framework has several names and formulations:

- **Predictive Processing / Predictive Coding** (Rao & Ballard, 1999; Friston, 2005)
- **The Free Energy Principle** (FEP) (Friston, 2006, 2010): A mathematical framework grounding PP in variational inference and information theory.
- **Active Inference** (Friston, 2009): The extension of PP to action, where organisms act on the world to fulfill their predictions.
- **The Bayesian Brain Hypothesis** (Knill & Pouget, 2004): The broader claim that neural computation implements Bayesian inference.
- **Controlled Hallucination** (Seth, 2021): A vivid description of PP's view of conscious perception.

1.2 Historical Context

1.2.1 Helmholtz and Unconscious Inference

The idea that perception involves inference dates to **Hermann von Helmholtz** (1867), who proposed that the brain performs "unconscious inferences" to interpret ambiguous sensory data. Helmholtz recognized that the retinal image is inherently ambiguous (multiple 3D scenes can produce the same 2D image) and that the brain must use prior knowledge to resolve this ambiguity.

1.2.2 The Computational Turn

In the late 20th century, several streams converged:

- **Bayesian statistics:** Formalized the combination of prior knowledge with new evidence.
- **Machine learning:** Generative models, variational methods, and hierarchical Bayesian models provided computational tools.
- **Neuroscience:** Discovery of massive feedback connections in cortex (feedback outnumbers feedforward ∼10:1 in many areas) demanded explanation.
- **Rao & Ballard (1999):** Demonstrated that predictive coding could explain extra-classical receptive field properties in V1 using a hierarchical model.

1.2.3 Friston and the Free Energy Principle

Karl Friston (University College London) unified these ideas under the **Free Energy Principle** (FEP), arguing that all adaptive behavior—perception, action, learning, development, evolution—can be understood as minimizing a single quantity: **variational free energy**. This transformed PP from a model of perception into a unified theory of brain function and, potentially, consciousness.

1.3 Overview of the Framework

1. **Generative model:** The brain encodes a hierarchical, probabilistic model of the causal structure of the world.
2. **Prediction:** Each level of the hierarchy generates predictions about the level below.
3. **Prediction error:** The difference between prediction and actual input is computed and propagated upward.
4. **Inference:** The brain minimizes prediction error by updating its model (perception) or by acting on the world (action).
5. **Precision:** The brain estimates the reliability (precision) of its predictions and prediction errors, and weights them accordingly (attention).
6. **Consciousness:** Arises from the content of the generative model—the brain's "best guess" about the world and itself.

Chapter 2

Mathematical Foundations

This chapter develops the mathematical apparatus underlying PP and the Free Energy Principle. The formalism is essential for understanding the theory's explanatory power and its connection to consciousness.

2.1 Bayesian Inference

PP is fundamentally Bayesian. Given observed data $\mathbf{y}$ and hidden causes $\boldsymbol{\theta}$:

Bayes' theorem:

$$\underbrace{p(\boldsymbol{\theta} \mid \mathbf{y})}_{\text{posterior}} = \frac{\overbrace{p(\mathbf{y} \mid \boldsymbol{\theta})}^{\text{likelihood}} \cdot \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\mathbf{y})}_{\text{evidence (marginal likelihood)}}}$$

- **Prior** $p(\boldsymbol{\theta})$: The brain's existing beliefs about the world before observing data.
- **Likelihood** $p(\mathbf{y} \mid \boldsymbol{\theta})$: How probable the sensory data are given a particular state of the world.
- **Posterior** $p(\boldsymbol{\theta} \mid \mathbf{y})$: The updated belief after incorporating the data.
- **Evidence** $p(\mathbf{y})$: The total probability of the data under the model (a normalizing constant).

Perception in PP is the process of computing (or approximating) the posterior $p(\boldsymbol{\theta} \mid \mathbf{y})$. The conscious percept corresponds to the posterior mode or mean—the brain's best estimate of the world's hidden causes.

2.2 The Problem of Intractability

Computing the exact posterior requires evaluating the evidence:

$$p(\mathbf{y}) = \int p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$$

For realistic generative models, this integral is **intractable**—it cannot be computed exactly because it involves summing over an astronomically large space of hidden causes. The brain must therefore use *approximate* inference.

2.3 Variational Inference

The Free Energy Principle uses **variational inference** to approximate the intractable posterior. The idea: instead of computing $p(\boldsymbol{\theta} \mid \mathbf{y})$ directly, find an approximate distribution $q(\boldsymbol{\theta})$ from a tractable family that is as close as possible to the true posterior.

Variational Free Energy

The **variational free energy** F is defined as:

$$F = \underbrace{D_{\mathrm{KL}}[q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})]}_{\geq 0 \text{ (KL divergence)}} - \underbrace{\ln p(\mathbf{y})}_{\text{log evidence}} \tag{2.1}$$

Rearranging:

$$F = \underbrace{-\mathbb{E}_q[\ln p(\mathbf{y} \mid \boldsymbol{\theta})]}_{\text{expected energy (prediction error)}} + \underbrace{D_{\mathrm{KL}}[q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta})]}_{\text{complexity (deviation from prior)}} \tag{2.2}$$

Equivalently:

$$F = -\underbrace{\mathbb{E}_q[\ln p(\mathbf{y}, \boldsymbol{\theta})]}_{\text{expected log joint}} + \underbrace{\mathbb{E}_q[\ln q(\boldsymbol{\theta})]}_{\text{negative entropy of } q} \tag{2.3}$$

Key Properties of Free Energy

1. $F \geq -\ln p(\mathbf{y})$. Free energy is an **upper bound** on surprise (negative log evidence). Minimizing F is equivalent to maximizing a lower bound on the evidence (the ELBO).
2. When $q(\boldsymbol{\theta}) = p(\boldsymbol{\theta} \mid \mathbf{y})$ (the approximate equals the true posterior), the KL divergence is zero and $F = -\ln p(\mathbf{y})$ exactly.
3. Minimizing F with respect to q performs **approximate Bayesian inference**: it brings q closer to the true posterior.
4. Minimizing F with respect to model parameters performs **learning**: it improves the generative model.
5. Minimizing F with respect to **actions** performs **active inference**: it changes sensory input to match predictions.

2.4 The Generative Model

At the heart of PP is a **hierarchical generative model** that specifies how hidden causes generate sensory observations.

Hierarchical Generative Model

A generative model consists of:

$$\text{Observation model:} \quad \mathbf{y} = g_0(\mathbf{x}_1) + \boldsymbol{\varepsilon}_0, \qquad \boldsymbol{\varepsilon}_0 \sim \mathcal{N}(0, \Pi_0^{-1}) \tag{2.4}$$

$$\text{Level } l \text{ dynamics:} \quad \mathbf{x}_l = g_l(\mathbf{x}_{l+1}) + \boldsymbol{\varepsilon}_l, \qquad \boldsymbol{\varepsilon}_l \sim \mathcal{N}(0, \Pi_l^{-1}) \tag{2.5}$$

$$\text{Top-level prior:} \quad \mathbf{x}_L \sim p(\mathbf{x}_L) \tag{2.6}$$

where $\mathbf{x}_l$ are hidden states at level l, g_l are nonlinear generative functions mapping higher levels to lower, $\boldsymbol{\varepsilon}_l$ are prediction errors (Gaussian noise), and Π_l are **precision matrices** (inverse covariance) encoding the reliability of predictions at each level.

The generative model specifies how the brain believes the world works: high-level causes ($\mathbf{x}_L$: objects, scenes, agents) generate intermediate representations ($\mathbf{x}_l$: surfaces, edges, motion) which

generate sensory data ($\mathbf{y}$: retinal images, cochlear signals).

2.5 Predictive Coding: Message Passing

Predictive coding implements variational inference in the hierarchical model through local message passing between cortical levels. At each level l:

Predictive Coding Update Equations

Prediction error:

$$\boldsymbol{\varepsilon}_l = \mathbf{x}_l - g_l(\hat{\mathbf{x}}_{l+1}) \tag{2.7}$$

State update (gradient descent on free energy):

$$\dot{\hat{\mathbf{x}}}_l = D\hat{\mathbf{x}}_l + \underbrace{\frac{\partial g_{l-1}^T}{\partial \hat{\mathbf{x}}_l}\,\Pi_{l-1}\,\boldsymbol{\varepsilon}_{l-1}}_{\text{bottom-up: prediction error from below}} - \underbrace{\Pi_l\,\boldsymbol{\varepsilon}_l}_{\text{top-down: prediction error at this level}} \tag{2.8}$$

Precision update:

$$\dot{\hat{\Pi}}_l \propto -\frac{\partial F}{\partial \hat{\Pi}_l} = \frac{1}{2}\left(\hat{\Pi}_l^{-1} - \boldsymbol{\varepsilon}_l \boldsymbol{\varepsilon}_l^T\right) \tag{2.9}$$

Neural interpretation:

- $\hat{\mathbf{x}}_l$: activity of **representation units** (deep-layer pyramidal neurons) encoding the estimated state.
- $\boldsymbol{\varepsilon}_l$: activity of **prediction error units** (superficial-layer pyramidal neurons).
- $g_l(\hat{\mathbf{x}}_{l+1})$: **feedback (top-down) predictions** sent from level $l+1$.
- $\Pi_l \boldsymbol{\varepsilon}_l$: **precision-weighted prediction error** sent feedforward to level $l+1$.
- Feedforward signals carry prediction errors; feedback signals carry predictions.

2.6 Precision and Attention

Precision Π_l is the inverse variance of prediction errors at level l. It encodes the **reliability** or **confidence** in signals:

Precision-weighted prediction error: $\xi_l = \Pi_l\,\boldsymbol{\varepsilon}_l$

High precision Π_l means prediction errors at level l are trusted and given high weight. Low precision means they are unreliable and down-weighted.

Precision ≡ Attention

In PP, **attention** is the optimization of precision. Attending to a stimulus means increasing the precision (gain) of prediction errors arising from that stimulus, so they have more influence on the posterior estimate. This connects directly to gain modulation in neural

circuits:

$$\text{Attention to level } l \;\equiv\; \text{increasing } \Pi_l \;\equiv\; \text{amplifying prediction errors at level } l$$

This provides a principled, mathematically grounded account of attention within the PP framework.

Chapter 3

The Free Energy Principle

3.1 From Predictive Coding to Universal Principle

Karl Friston elevated predictive coding from a model of perception to a **universal principle of self-organizing systems**. The Free Energy Principle (FEP) states:

The Free Energy Principle

Any self-organizing system that maintains its structural and functional integrity must minimize its **variational free energy**. This is equivalent to:

1. Minimizing **surprise** (negative log evidence) about sensory states
2. Maximizing the **evidence** for the system's generative model of the world
3. Performing approximate **Bayesian inference** on the causes of sensory input
4. Maintaining the system within a bounded set of states (its "attracting set")

Formally: biological systems minimize $F = \mathbb{E}_q[\ln q(\boldsymbol{\theta}) - \ln p(\mathbf{y}, \boldsymbol{\theta})]$.

3.2 Markov Blankets

A central concept in the FEP is the **Markov blanket**:

Markov Blanket

A **Markov blanket** of a set of states $\boldsymbol{\mu}$ (internal states) is the set of states $\mathbf{b}$ that separates $\boldsymbol{\mu}$ from everything else (external states $\boldsymbol{\eta}$), such that $\boldsymbol{\mu}$ and $\boldsymbol{\eta}$ are conditionally independent given $\mathbf{b}$:

$$p(\boldsymbol{\mu}, \boldsymbol{\eta} \mid \mathbf{b}) = p(\boldsymbol{\mu} \mid \mathbf{b})\, p(\boldsymbol{\eta} \mid \mathbf{b})$$

The blanket is partitioned into **sensory states s** (influenced by external states) and **active states a** (influencing external states):

$$\mathbf{b} = (\mathbf{s}, \mathbf{a})$$

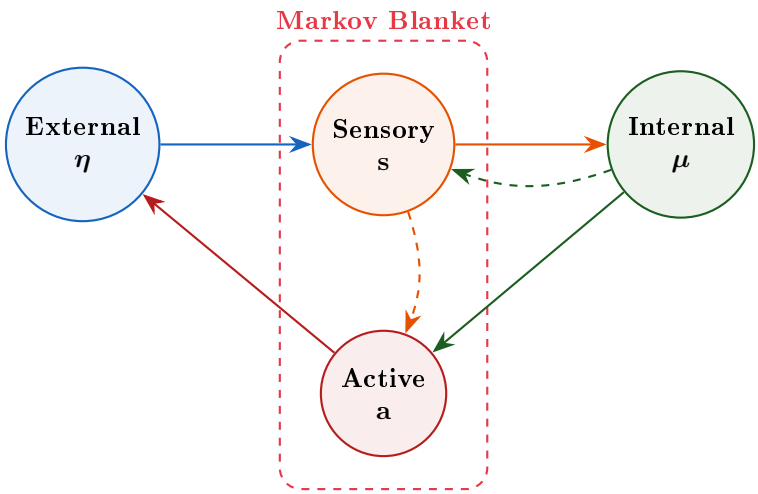

Figure: Markov blanket separating internal states from external states. Sensory states mediate inward influence; active states mediate outward influence.

For a brain: external states are the world; sensory states are receptor activations; active states are motor commands; internal states are neural activity encoding beliefs about the world.

3.3 Surprise and Self-Evidencing

The FEP can be understood through the concept of **surprise** (or **surprisal**):

$$\text{Surprise:} \quad \mathfrak{S}(\mathbf{y}) = -\ln p(\mathbf{y} \mid m)$$

where m is the organism's generative model. High surprise means the sensory input was unexpected under the model. The free energy $F \geq \mathfrak{S}$ provides an upper bound on surprise.

An organism that minimizes free energy over time will, on average, occupy states with low surprise—states that are expected under its model. Since the model encodes the organism's "preferred" states (e.g., a fish expects to be in water, at a certain temperature, etc.), minimizing surprise is equivalent to maintaining the organism within its viable states. Friston calls this **self-evidencing**: the organism provides evidence for its own model of existence.

3.4 Active Inference

The FEP extends to action through **active inference**: instead of only updating beliefs to match sensory input (perception), the organism can also change the world to match its predictions (action).

Active Inference

Free energy can be minimized in two ways:

1. **Perceptual inference:** Update internal states $\boldsymbol{\mu}$ (beliefs) to better predict sensory input:
$$\dot{\boldsymbol{\mu}} = -\frac{\partial F}{\partial \boldsymbol{\mu}} \quad \text{(gradient descent on } F \text{ w.r.t. beliefs)}$$
2. **Active inference:** Update active states $\mathbf{a}$ (actions) to change sensory input to match

predictions:

$$\dot{\mathbf{a}} = -\frac{\partial F}{\partial \mathbf{a}} = -\frac{\partial \mathbf{y}}{\partial \mathbf{a}} \cdot \frac{\partial F}{\partial \mathbf{y}} \quad \text{(gradient descent on } F \text{ w.r.t. actions)}$$

Action is driven by the same quantity as perception—prediction error—but instead of updating the model, it changes the world.

Active Inference: Picking Up a Cup

You predict the sensation of holding a cup (proprioceptive predictions of arm position, tactile predictions of cup contact). The prediction error (arm is not yet holding cup) drives motor commands that move the arm to grasp the cup. When the cup is grasped, prediction errors are minimized. You did not need a separate "motor plan"—the predictions themselves drive action.

3.5 Expected Free Energy and Planning

For planning and decision-making, active inference uses **expected free energy** G:

$$G(\pi) = \underbrace{\mathbb{E}_q[D_{\mathrm{KL}}[q(\boldsymbol{\theta} \mid \mathbf{y}_\pi) \parallel q(\boldsymbol{\theta})]]}_{\text{epistemic value (information gain)}} + \underbrace{\mathbb{E}_q[-\ln p(\mathbf{y}_\pi)]}_{\text{pragmatic value (expected surprise)}}$$

where π denotes a policy (sequence of actions) and $\mathbf{y}_\pi$ denotes the predicted sensory outcomes under policy π. The brain selects the policy π^* that minimizes expected free energy:

$$\pi^* = \arg\min_\pi G(\pi)$$

This decomposes into two drives: **epistemic value** (seeking information, resolving uncertainty) and **pragmatic value** (achieving preferred outcomes). The balance between exploration and exploitation emerges naturally.

Chapter 4

Hierarchical Predictive Coding in the Brain

4.1 Cortical Microcircuit Architecture

Predictive coding maps naturally onto the **laminar structure** of cortex:

Cortical Implementation of Predictive Coding

- **Deep pyramidal neurons** (layers 5/6): Encode expectations $\hat{\mathbf{x}}_l$ (representation units). Send top-down predictions via feedback connections to the level below.
- **Superficial pyramidal neurons** (layer 2/3): Compute and encode prediction errors $\boldsymbol{\varepsilon}_l$. Send prediction errors feedforward to the level above.
- **Inhibitory interneurons** (layers 2/3, 4): Mediate the subtraction operation (actual input minus prediction) and regulate gain (precision).
- **Feedback connections** (layers 5/6 → layers 1, 5/6): Carry predictions from higher to lower areas.
- **Feedforward connections** (layer 2/3 → layer 4): Carry prediction errors from lower to higher areas.

4.2 Oscillatory Signatures

Predictive coding maps onto specific oscillatory frequency bands:

Signal	Frequency Band	Function in PP
Predictions (top-down)	Alpha/beta (8–30 Hz)	Carry predictions from higher to lower areas
Prediction errors (bottom-up)	Gamma (>30 Hz)	Carry precision-weighted errors upward
Precision modulation	Alpha (8–13 Hz)	Inhibitory gating; low alpha = high precision
Cross-frequency coupling	Theta-gamma nesting	Temporal coordination of hierarchical inference

This pattern—gamma for feedforward prediction errors, alpha/beta for feedback predictions—has been confirmed by van Kerkoerle et al. (2014) and Bastos et al. (2015), as discussed in the RPT chapter.

4.3 Empirical Evidence for Predictive Coding

4.3.1 Mismatch Negativity (MMN)

The **mismatch negativity** is an event-related potential (∼100–250 ms) generated when an auditory stimulus violates a regular pattern (e.g., an occasional deviant tone in a sequence of standard tones). PP explains MMN directly: the brain builds a predictive model of the auditory sequence; the deviant triggers a large prediction error.

The MMN amplitude scales with the **precision-weighted prediction error**—deviants that are more surprising (less expected) or more reliable (in quiet conditions) produce larger MMN.

4.3.2 Repetition Suppression and Expectation Suppression

Repetition suppression: Repeated stimuli evoke progressively weaker neural responses. PP interpretation: the generative model learns to predict the repeated stimulus, reducing prediction error.

Expectation suppression: Expected stimuli (even novel ones) evoke weaker responses than unexpected stimuli. PP interpretation: prediction errors are small when the stimulus matches the model's prediction.

4.3.3 Perceptual Illusions

PP naturally explains illusions as cases where the generative model's prior dominates over weak sensory evidence:

The Hollow Mask Illusion

A hollow (concave) face mask appears convex—the brain "sees" a normal face. PP explanation: the prior for faces (faces are convex) is so strong that it overrides the (relatively weak) sensory evidence for concavity. The generative model's prediction (convex face) wins over the data.

This illusion is disrupted in schizophrenia—consistent with the hypothesis that schizophrenia involves weakened priors (see Section 5.4).

Chapter 5

Consciousness in Predictive Processing

PP is not a theory of consciousness per se, but several researchers have developed explicit accounts of consciousness within the PP framework.

5.1 Consciousness as the Content of the Generative Model

The simplest PP account: **conscious experience is the content of the brain's generative model**—the posterior estimate $\hat{\mathbf{x}}$ at all levels of the hierarchy.

Consciousness as Bayesian Best Guess

What we consciously experience is not the raw sensory data $\mathbf{y}$, but the brain's **posterior estimate** of the hidden causes of that data:

$$\text{Conscious percept} \approx \hat{\mathbf{x}} = \arg\max_{\mathbf{x}} \; p(\mathbf{x} \mid \mathbf{y}) \approx \arg\min_{\mathbf{x}} \; F(\mathbf{x}, \mathbf{y})$$

Perception is a "controlled hallucination": the brain generates a model of the world and constrains it with sensory data. What we see, hear, and feel is the model's output, not the data itself.

5.1.1 "Controlled Hallucination" (Anil Seth)

Anil Seth (University of Sussex) has developed the most vivid characterization: conscious perception is a **controlled hallucination**. The brain is always hallucinating—generating a model of the world from the top down. Sensory data merely *constrains* the hallucination, keeping it consistent with reality.

When the hallucination is well-constrained by data, we call it "perception." When it is poorly constrained (as in dreaming, psychedelics, or psychosis), the hallucination becomes more vivid, bizarre, or dissociated from reality.

5.2 Seth's "Beast Machine" Framework

Seth (2021) extends the PP account specifically to consciousness, proposing:

Seth's Predictive Processing Account of Consciousness

1. **Perceptual consciousness** = the content of the generative model's posterior estimates about the external world (colors, shapes, sounds).
2. **Self-consciousness / embodied selfhood** = the content of the generative model's posterior estimates about the *body* (interoception, proprioception, agency).
3. **Emotional experience** = precision-weighted interoceptive prediction errors integrated

with allostatic regulation.

4. **The "real problem" of consciousness:** Rather than asking *why* there is experience (the Hard Problem), Seth proposes the *real problem*: explaining the specific character and structure of experience in terms of generative models.

5.3 Precision, Attention, and the "Level" of Consciousness

A key PP contribution is the account of how the **level** (quantity) of consciousness relates to precision:

The overall "level" of consciousness $\mathcal{L}$ may be related to the total precision-weighted prediction error across the hierarchy:

$$\mathcal{L} \propto \sum_l \mathrm{tr}\left(\Pi_l \, \boldsymbol{\varepsilon}_l \boldsymbol{\varepsilon}_l^T\right) \quad \text{or} \quad \mathcal{L} \propto \sum_l \mathrm{rank}(\Pi_l)$$

When precision collapses (as in anesthesia, deep sleep), prediction errors are not weighted and the generative model is not updated—consciousness fades.

Anesthesia in PP terms: anesthetic agents disrupt the precision-weighting of prediction errors, preventing the hierarchical inference that constitutes conscious perception. Sensory cortex may still generate local responses (feedforward), but without precision-weighted feedback, the generative model cannot be updated, and consciousness is lost.

5.4 Psychedelics, Schizophrenia, and Altered States

PP provides elegant accounts of altered states of consciousness:

5.4.1 Psychedelics: The REBUS Model

Carhart-Harris & Friston (2019) proposed the **REBUS** model (**R**elaxed **B**eliefs **U**nder p**S**ychedelics):

REBUS Model

Psychedelics (e.g., psilocybin, LSD) relax the precision of high-level priors—the deep beliefs that normally constrain perception. This has several consequences:

1. Bottom-up prediction errors gain more influence relative to top-down predictions.
2. Perception becomes more driven by sensory data and less by prior expectations.
3. The generative model enters a state of increased entropy (more possible perceptual configurations).
4. This produces the characteristic psychedelic phenomenology: vivid colors, geometric patterns, synesthesia, ego dissolution, mystical experiences.

Formally: psychedelics reduce Π_l at high levels l, increasing the **relative precision** of lower-level prediction errors.

5.4.2 Schizophrenia: Aberrant Precision

PP accounts of schizophrenia (Adams et al., 2013; Sterzer et al., 2018) propose that psychotic symptoms arise from **aberrant precision estimation**:

- **Hallucinations:** Excessive precision on priors (strong predictions not constrained by data) → the generative model generates percepts that are not driven by sensory evidence.
- **Delusions:** Excessive precision on prediction errors at high levels (overweighting unexpected events) → the model is continually revised in response to noise, leading to unstable, conspiracy-like beliefs.
- **Disrupted MMN:** Reduced precision of sensory prediction errors → weaker mismatch responses, consistent with empirical findings.

5.4.3 Dreaming

During dreaming, sensory input is largely absent. The generative model runs "offline," generating predictions without sensory constraint. PP predicts:

- Dream content reflects the generative model's priors and predictions (hence familiar themes, locations, people).
- Dreams are bizarre because high-level priors are partially relaxed (similar to psychedelics).
- Lucid dreaming may involve increased precision of "meta-cognitive" prediction errors (awareness that one is dreaming).

5.5 Interoceptive Inference and Emotional Consciousness

Seth and Friston have extended PP to **interoception**—the brain's perception of its own body:

Interoceptive Predictive Coding

The brain maintains a generative model of the body's internal states (heart rate, blood pressure, gut motility, etc.). Interoceptive prediction errors drive:

1. **Interoceptive inference:** Updating beliefs about internal body states ("my heart is beating fast").
2. **Active interoceptive inference:** Autonomic regulation that changes body states to match predictions (e.g., increasing heart rate to match predicted exertion).
3. **Emotional experience:** Emotions are the conscious content of interoceptive inference—the brain's best guess about the causes of its interoceptive signals.

This connects to the James–Lange theory of emotion and to Lisa Feldman Barrett's **theory of constructed emotion**: emotions are constructed by the brain's predictions about interoceptive states.

5.6 The Self in Predictive Processing

PP provides a naturalistic account of the **self**:

The Predictive Self

The "self" is not a fixed entity but a **multilayered generative model**:

1. **Bodily self:** The generative model's estimates of body state (proprioception, interoception, body schema). Disruption → depersonalization, out-of-body experiences.
2. **Perspectival self:** The model's first-person viewpoint (embodied in the body schema and spatial frame). Disruption → autoscopy.

3. **Volitional self:** The model's predictions about the consequences of its own actions (sense of agency). Disruption → passivity symptoms in schizophrenia.
4. **Narrative self:** High-level, temporally extended model of personal history and identity. Disruption → dissociative disorders.
5. **Social self:** The model's representation of how others model the self (higher-order social inference).

Chapter 6

Formal Extensions and Advanced Topics

6.1 Generalized Coordinates of Motion

Friston's formulation uses **generalized coordinates**—not just the state $\mathbf{x}$ but also its temporal derivatives:

$$\tilde{\mathbf{x}} = (\mathbf{x},\ \mathbf{x}',\ \mathbf{x}'',\ \ldots) = \left(\mathbf{x},\ \frac{d\mathbf{x}}{dt},\ \frac{d^2\mathbf{x}}{dt^2},\ \ldots\right)$$

The free energy is then defined over generalized coordinates $\tilde{\mathbf{x}}$, and the generative model specifies dynamics at all temporal orders. This allows the brain to model not just the current state but its velocity, acceleration, and higher-order dynamics—essential for predicting smooth, temporally extended processes.

6.2 Dynamic Causal Modeling (DCM)

Dynamic Causal Modeling is Friston's framework for inferring effective connectivity between brain regions from neuroimaging data:

Neural state equation:

$$\dot{\mathbf{z}} = (A + \sum_j u_j B_j)\,\mathbf{z} + C\,\mathbf{u}$$

where $\mathbf{z}$ are neural states, A is the intrinsic connectivity matrix, B_j encodes how experimental input u_j modulates connectivity, and C maps external inputs. DCM performs Bayesian model comparison to determine which connectivity structure best explains the data—embodying the FEP at the meta-level.

6.3 Markov Blankets and Consciousness

Friston and colleagues have explored a deep connection between Markov blankets and conscious selfhood:

Markov Blanket → Self-Model → Consciousness

1. Any system with a Markov blanket can be described *as if* it is performing inference on external states.
2. Systems with **deep** (hierarchical) internal structure can perform sophisticated inference.
3. If the system's generative model includes a model of *its own Markov blanket* (the boundary between self and world), it possesses a form of **self-model**.
4. Consciousness may require a system whose generative model includes a model of itself *as a system performing inference*—a kind of meta-model or "strange loop."

This connects FEP to the philosophical concept of self-consciousness and to Hofstadter's

"strange loops."

6.4 Renormalization Group and Hierarchical Depth

Recent work connects the FEP to **renormalization group** (RG) methods from physics:

In statistical physics, the RG describes how the effective description of a system changes across spatial/temporal scales. In PP:

$$\mathbf{x}_{l+1} = \mathcal{R}(\mathbf{x}_l)$$

where $\mathcal{R}$ is a coarse-graining operation that maps detailed states at level l to summary statistics at level $l+1$. Each level of the cortical hierarchy implements a renormalization step, extracting progressively more abstract features. The *depth* of the hierarchy may relate to the *richness* of conscious experience: deeper models $\rightarrow$ more differentiated, abstract consciousness.

6.5 The Path Integral Formulation

The most general formulation of the FEP uses **path integrals**:

The probability of a particular trajectory (path) $\tilde{\mathbf{x}}[0,T]$ of internal states:

$$p(\tilde{\mathbf{x}}[0,T]) = \frac{1}{Z}\exp\left(-\int_0^T \mathcal{L}\left(\tilde{\mathbf{x}}(t), \dot{\tilde{\mathbf{x}}}(t)\right) dt\right)$$

where $\mathcal{L}$ is a Lagrangian that includes both the free energy and a kinetic term. The most probable path (the one that dominates the integral) is the one that minimizes the action:

$$S[\tilde{\mathbf{x}}] = \int_0^T \mathcal{L}\left(\tilde{\mathbf{x}}, \dot{\tilde{\mathbf{x}}}\right) dt$$

This connects the FEP to the principle of least action in physics, suggesting a deep formal analogy between self-organizing biological systems and physical systems.

Chapter 7

PP vs. Other Theories of Consciousness

7.1 PP vs. Global Workspace Theory

Key Relationships: PP vs. GWT

Dimension	PP / Active Inference	GWT / GNWT
What is consciousness?	Content of generative model	Globally broadcast information
Mechanism	Hierarchical Bayesian inference	Ignition + broadcast
Attention	Precision optimization	Competition for workspace
Role of PFC	High-level prior / control	Workspace hub
Compatibility	PP could describe *how* the workspace works; GWT describes *what* it does Key difference	PP is a computational framework; GWT is an architectural claim

PP and GWT are often viewed as **complementary**: PP describes the *algorithm* (Bayesian inference via prediction error minimization); GWT describes the *architecture* (global workspace with ignition and broadcast). The global workspace could be the mechanism by which high-level predictions are broadcast and prediction errors are propagated.

7.2 PP vs. Integrated Information Theory

Key Differences: PP vs. IIT

Dimension	PP / Active Inference	IIT
Starting point	Computational / normative	Phenomenological axioms
Consciousness is...	Content of Bayesian inference	Integrated information (Φ)
Hard Problem	The "real problem" (Seth)	Identity claim
Panpsychism	Not necessarily (requires deep model)	Yes ($\Phi > 0$)
Brain location	Hierarchical cortex	Posterior cortex
Key measure	Free energy / prediction error	Φ
Overlap	Φ could measure the complexity/integration of the generative model	

7.3 PP vs. Recurrent Processing Theory

PP and RPT are highly compatible: RPT's "recurrent processing" is precisely the iterative exchange of predictions and prediction errors in predictive coding. RPT claims recurrence is *necessary* for consciousness; PP specifies *what* the recurrent processing computes (Bayesian inference). The convergence is strong: both identify feedback processing as essential and feedforward processing as insufficient.

7.4 PP vs. Higher-Order Theories

PP's precision-weighted metacognitive readout connects to HOT: the higher-order representation could *be* the generative model's estimate of its own perceptual states. Meta-d' (the higher-order sensitivity measure from HOT) could reflect the precision of metacognitive prediction errors within the PP hierarchy.

7.5 PP vs. Attention Schema Theory

AST's "attention schema" is structurally similar to PP's generative model of its own attentional (precision-optimization) process. Both claim that consciousness involves self-modeling; PP provides the computational framework (free energy minimization); AST provides the specific content (a model of attention).

Chapter 8

Criticisms and Open Questions

8.1 Common Critiques

Critique 1: The FEP is unfalsifiable.

The FEP is sometimes accused of being so general that it cannot be falsified: any behavior of any self-organizing system can be redescribed as free energy minimization.

Response: The FEP is best understood as a *framework* (like Newtonian mechanics), not a specific hypothesis. It becomes falsifiable when instantiated in specific models (specific generative models, specific neural implementations). Predictive coding as implemented in DCM makes specific, testable predictions.

Critique 2: PP doesn't explain consciousness—it explains information processing.

Even if the brain performs Bayesian inference, why does this inference feel like something? The Hard Problem persists.

Response: Seth advocates the "real problem" approach: rather than asking why there is experience, explain the *structure* of experience. PP makes specific predictions about what determines the content, quality, and level of consciousness. Whether this dissolves or merely side-steps the Hard Problem is debated.

Critique 3: The brain may not actually do Bayesian inference.

Neural circuits may use heuristics, approximations, or entirely non-Bayesian algorithms. The "Bayesian brain" may be an idealization.

Response: PP proponents agree that the brain performs *approximate* inference. The claim is not that neurons literally compute Bayes' theorem, but that the brain's computations can be usefully described as approximate Bayesian inference, just as physical systems can be described as minimizing energy even though particles don't "calculate" energies.

Critique 4: Precision is doing too much work.

In PP, precision explains attention, consciousness level, psychopathology, and more. It may be that the concept is overloaded and vague.

Response: Precision has specific mathematical meaning (inverse variance) and makes specific predictions (e.g., about gain modulation, oscillatory dynamics, and pharmacological effects). However, more empirical work is needed to test whether a single mechanism accounts for all these phenomena.

Critique 5: Consciousness in PP is epiphenomenal.

If all the work is done by prediction error minimization (an unconscious computational process), consciousness seems like a bystander.

Response: In PP, consciousness is not epiphenomenal—it *is* the content of the generative model

that drives behavior. The model's content causally influences actions (via active inference). Whether this addresses the deeper concern about the causal role of phenomenal experience depends on one's philosophical commitments.

8.2 Open Questions

1. **What distinguishes conscious from unconscious inference?** The brain performs many inferences unconsciously. What makes some inferences conscious? Is it precision, hierarchical depth, temporal persistence, or something else?
2. **Can PP explain the unity of consciousness?** How do multiple generative models (visual, auditory, bodily) combine into a unified experience?
3. **What is the role of temporal depth?** Consciousness may require inference over extended time scales (not just instantaneous states). How does the brain implement temporal depth?
4. **How does PP account for the neural correlates of consciousness?** Specific mapping from PP quantities to empirical signatures (P3b, gamma, PCI) is still developing.
5. **Is the FEP sufficient for consciousness?** Or is additional architectural structure (like a global workspace) needed?
6. **Artificial consciousness:** Would a machine implementing PP be conscious? If so, what kind of generative model would suffice?
7. **Can PP account for phenomenal richness?** The richness of experience seems to exceed what a finite generative model can encode.

Chapter 9

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